

Co-expression, cell-cell communication, and gene regulatory network analysis in the aging mouse heart

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1 Introduction

Cardiac aging involves coordinated cellular and molecular changes—such as fibrosis and metabolic shifts—that increase cardiovascular disease risk. While single-cell RNA sequencing (scRNA-seq) resolves cell heterogeneity, a single dataset alone leaves three questions unanswered: which genes co-vary (co-expression), who signals to whom (cell-cell communication), and what regulates what (transcription factors). Answering all three from the same cells provides convergent, cross-validated computational evidence for candidate genes.

hdWGCNA forms the study’s backbone by identifying cell-type-specific, age-associated co-expression modules and hub genes while mitigating scRNA-seq dropout. Three complementary tools address the remaining layers: LIANA+ (with Tensor-cell2cell) descriptively ranks consensus ligand-receptor pairs and latent age trajectories; MultiNicheNet inferentially prioritizes plausible signaling contrast between young and old states; and pySCENIC reconstructs transcription factor regulons to pinpoint drivers of these expression shifts.

Concretely, this study aims to: (1) identify age-associated gene co-expression modules in the aging mouse heart; (2) characterize key hub genes via GO/KEGG/Reactome enrichment; (3) test for overlap between age-associated genes, signaling partners (LIANA+, MultiNicheNet), and regulons (SCENIC); and (4) provide multi-layered, convergent evidence for the genes, cell types, and pathways driving cardiac aging.

All four analyses are run on the same upstream object (same cells, same `final_celltype` annotation, same `sample_id`/age structure), and the same WGCNA outputs (hub genes, age-associated modules) are used as the cross-validation reference throughout (Figure 1).

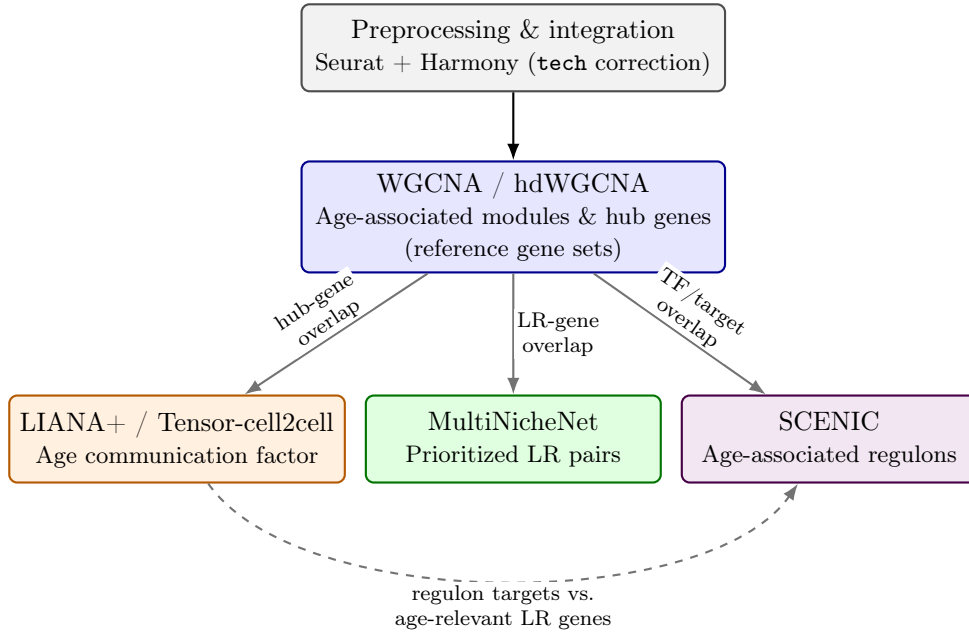


Figure 1: Overview of the four-layer pipeline. Solid arrows indicate the shared upstream data flow; double-headed arrows indicate bidirectional cross-validation, where WGCNA/hdWGCNA age-associated modules and hub genes serve as the common reference gene set for the other three analyses, and LIANA+ and SCENIC are additionally cross-checked against each other.

2 Data

Single-cell RNA-seq data from mouse heart tissue were obtained from the *Tabula Muris Senis* project [3], downloaded from Figshare as two independent archives corresponding to two sequencing technologies:

- **heart_facs**: cells profiled with plate-based, full-length Smart-seq2 (FACS-sorted).
- **heart_droplet**: cells profiled with droplet-based 10x Genomics chemistry.

Both archives include metadata for donor mouse identity, sex, chronological age, and cell ontology class (cell-type annotation). Across the two archives combined, 24 mice are represented, spanning six age points from 1 to 30 months (Table 1).

Age (months)	male	female	Total
1	1	0	1
3	5	5	10
18	2	1	3
21	0	2	2
24	6	0	6
30	2	0	2
Total	16	8	24

Table 1: Number of mice per sex and age point in the study.

3 Preliminary data visualization

This section presents the exploratory analysis, integration, quality control, and cell-type annotation of the raw FACS and droplet archives described in Section 2, prior to running the four analyses in Section 4.

3.1 Per-technology exploratory analysis

Prior to integration, the sex and age composition of each archive was examined independently (Figures 2, 3; Table 2), to confirm representation across experimental groups before combining datasets. Age coverage differs by technology: droplet spans all six age points, while FACS covers only the 3- and 24-month points.

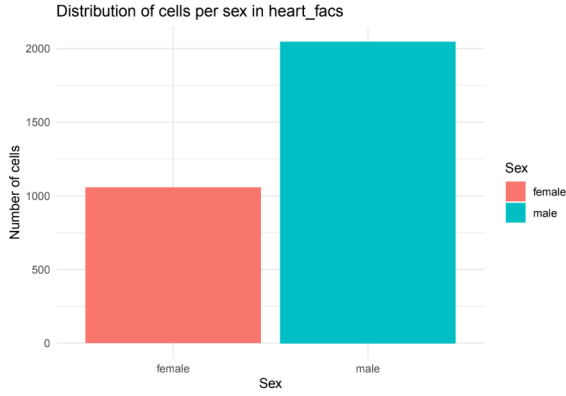


Figure 2: Bar plot of cell counts per sex for heart_facs

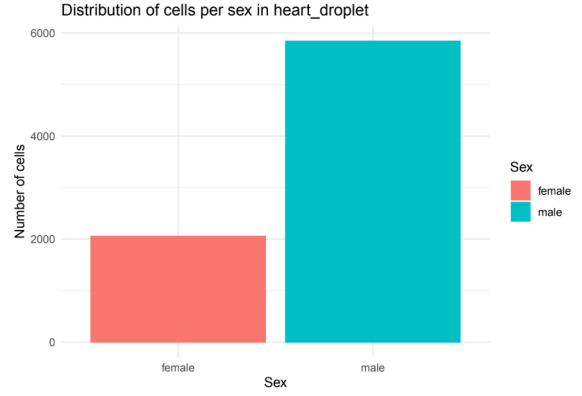


Figure 3: Bar plot of cell counts per sex for heart_droplet

Age	female	male
1m	0	842
3m	520	0
18m	637	486
21m	908	0
24m	0	744
30m	0	3779

(a) Heart Droplet

Age	female	male
3m	1057	1480
24m	0	567

(b) Heart FACS

Table 2: Age $\times$ sex cross-tabulation (cell counts) for heart droplet and heart FACS datasets.

3.2 Dataset integration

Gene symbols were used as the common identifier between the two archives. The shared gene set (intersection of `heart_facs` and `heart_droplet` gene panels) was retained, and the two `AnnData` objects were concatenated along the cell axis (outer join on genes), yielding a combined dataset labeled by technology (`tech`: *facs* / *droplet*). Gene symbol collisions introduced by character sanitization (underscore-to-hyphen conversion) were resolved with unique suffixes to guarantee a one-to-one mapping between gene identifiers and matrix rows.

A composite sample identifier, `sample_id`, was constructed by concatenating technology and donor mouse identity (or, when mouse identity was unavailable, technology, sex, and age). This identifier is used throughout the analysis to group cells by biological replicate (individual mouse) rather than by cell. The combined count matrix and metadata were assembled into a single `Seurat` object [4].

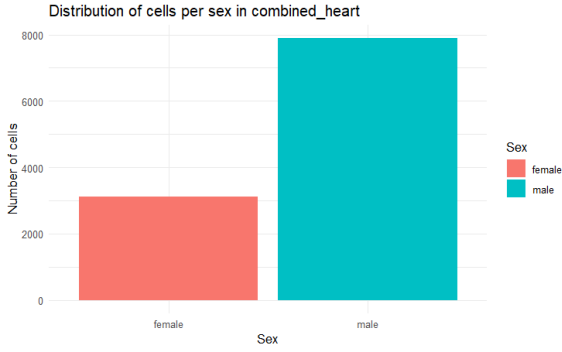


Figure 4: Bar plot of cell counts per sex for the combined dataset

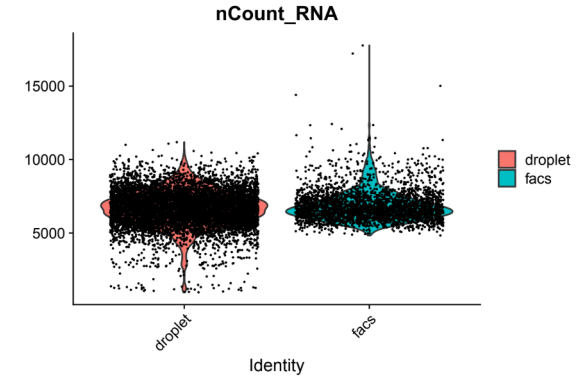


Figure 5: Comparison of total RNA counts (nCount_RNA) per cell between droplet-based and FACS-based sequencing technologies.

Age	female	male
1m	0	842
3m	1,577	1,480
18m	637	486
21m	908	0
24m	0	1,311
30m	0	3,779
Total	3,122	7,898

Table 3: Age $\times$ sex cross-tabulation cell counts for the combined dataset.

Metric	Count
Shared genes	19,473
Combined genes	21,577
Combined cells	11,020

(a) Combined dataset metrics

Technology (tech)	Cells
droplet	7,916
facs	3,104
Total	11,020

(b) Cells per sequencing technology

Table 4: Summary of gene overlap and cell counts across technologies.

3.3 Quality control and filtering

Cells were retained if they had more than 200 detected genes ($\text{nFeature_RNA} > 200$). This filter was applied jointly across both technologies rather than separately, because FACS (Smart-seq) libraries have substantially higher per-cell sequencing depth than droplet libraries; applying a shared threshold avoids introducing an artificial dependency between “genes detected” and sequencing technology that would otherwise confound the gene-selection step of network construction. Genes detected in fewer than 3 cells (globally) were removed.

3.4 Normalization and dimensionality reduction

Filtered counts were log-normalized, followed by identification of highly variable genes, feature scaling, and principal component analysis (PCA), all performed with standard **Seurat** functions.

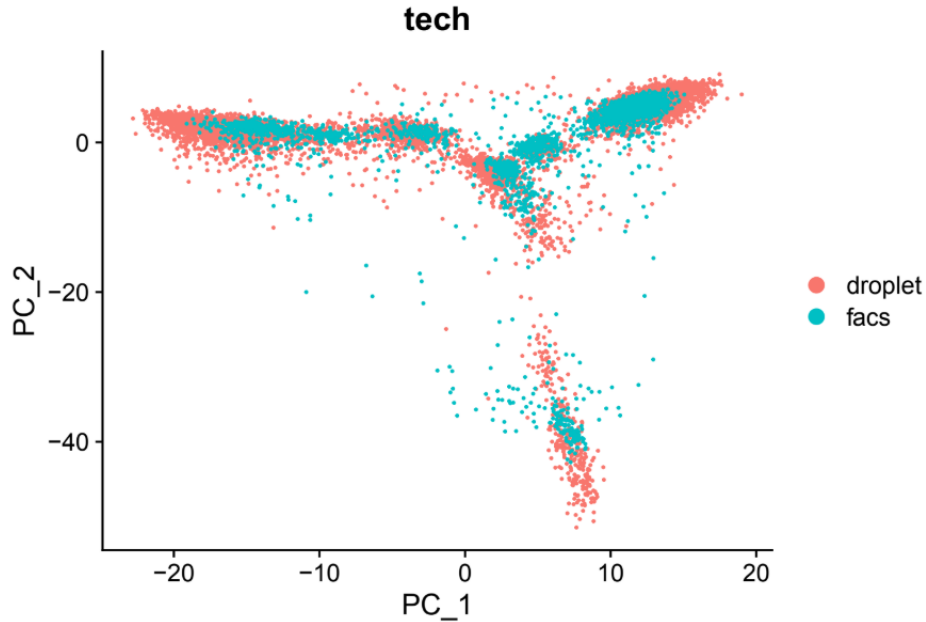


Figure 6: Principal component analysis, colored by sequencing technology (**tech**), before batch correction.

3.5 Batch integration across sequencing technologies

Cells were integrated across technologies using Harmony [5], run on the PCA embedding with **tech** as the batch covariate. A shared nearest-neighbor graph and UMAP embedding were computed on the Harmony-corrected components (20 and 15 dimensions, respectively).

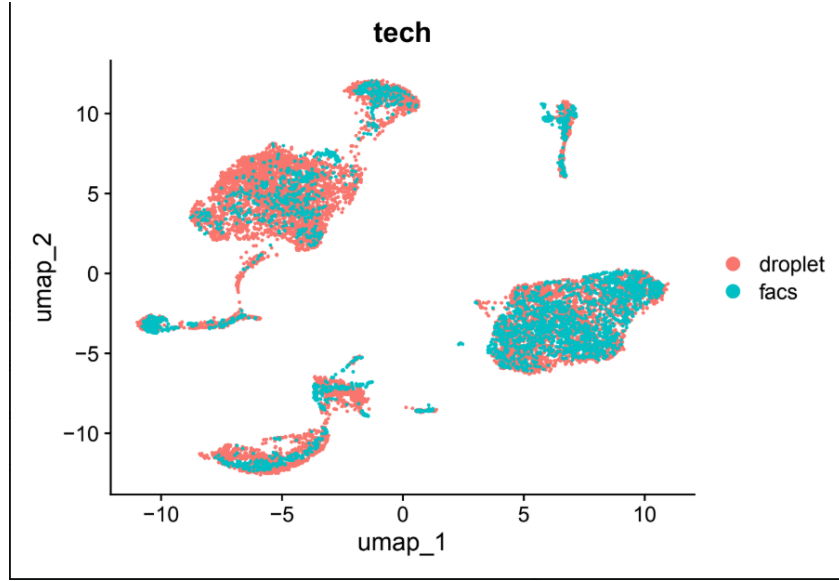


Figure 7: UMAP embedding colored by sequencing technology, after Harmony correction.

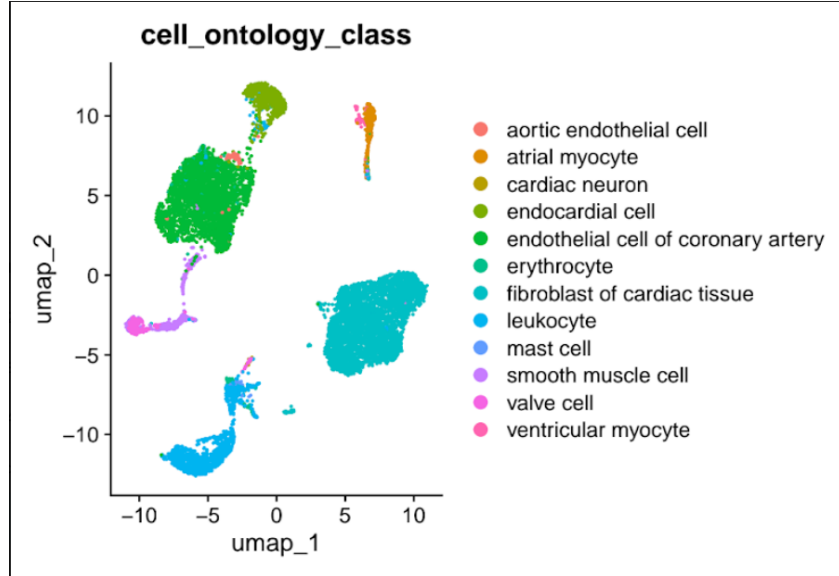


Figure 8: UMAP embedding colored by cell ontology class.

3.6 Cell type annotation

Cells lacking a valid cell ontology class annotation ("nan") were removed from further analysis. The resulting dataset defines the cell-type groups (`cell_ontology_class`) later collapsed into the eight `final_celltype` categories used throughout the co-expression and downstream analyses (Section 3.7).

Cell Type	Number of Cells
aortic endothelial cell	59
atrial myocyte	396
cardiac neuron	44
endocardial cell	708
endothelial cell of coronary artery	3364
erythrocyte	121
fibroblast of cardiac tissue	4041
leukocyte	1286
mast cell	57
smooth muscle cell	418
valve cell	204
ventricular myocyte	58

Table 5: Cell count distribution by ontology class (after NA removal). See Section 3.7 for how these are mapped to the eight `final_celltype` categories (mast cells and valve cells are excluded).

Cell Type	1m	3m	18m	21m	24m	30m
aortic endothelial cell	0	59	0	0	0	0
atrial myocyte	46	91	42	70	79	68
cardiac neuron	15	2	6	5	0	16
endocardial cell	30	190	112	118	137	121
endothelial cell of coronary artery	310	469	280	230	328	1,747
erythrocyte	2	1	10	10	2	96
fibroblast of cardiac tissue	261	1,634	406	288	457	995
leukocyte	86	333	135	89	183	460
mast cell	0	0	25	–	0	32
smooth muscle cell	88	56	33	23	36	182
valve cell	0	164	0	0	40	0
ventricular myocyte	0	38	–	0	20	0

Table 6: Cell type counts (n) across mouse ages (months).

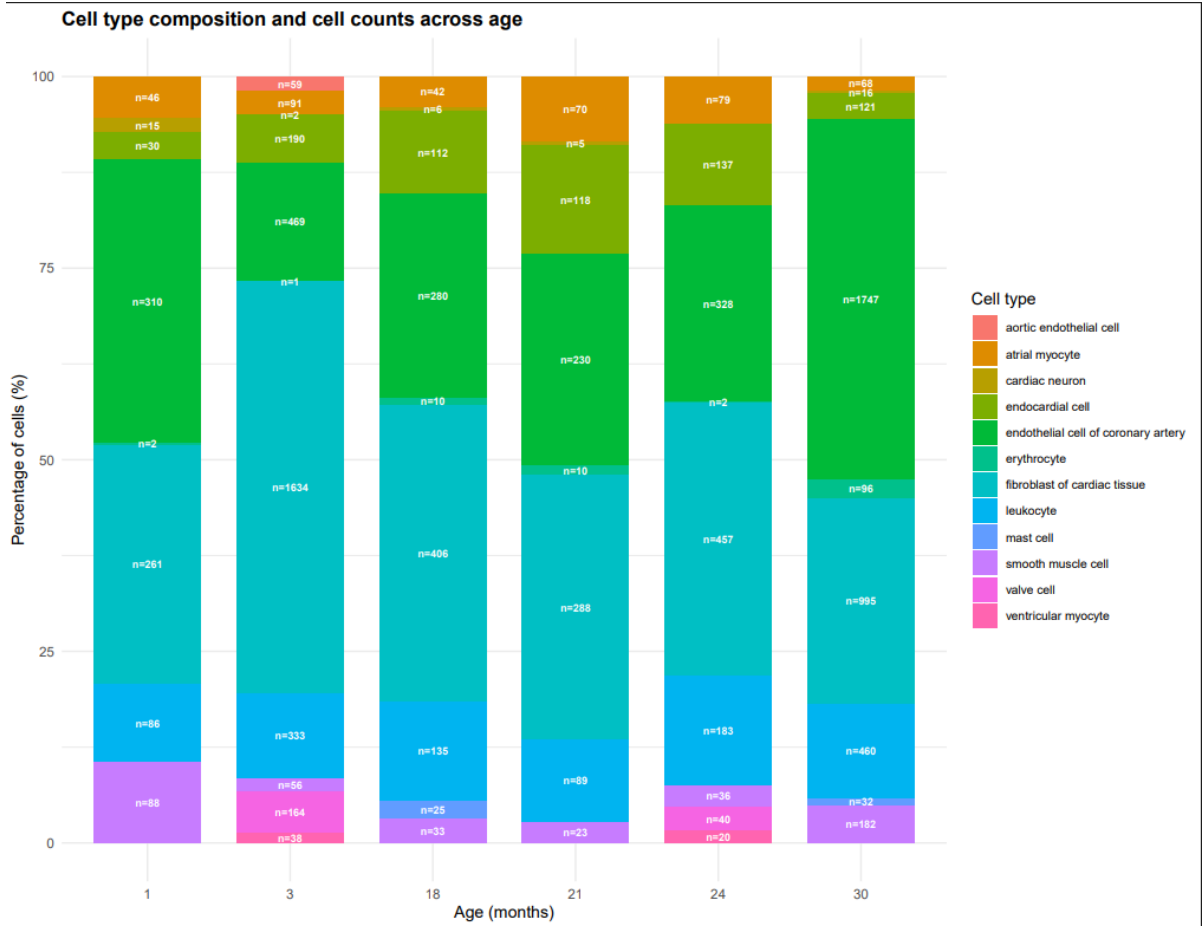


Figure 9: Number of cells per cell ontology class and age point.

3.7 Cell-type mapping to final categories

The raw `cell_ontology_class` labels (Table 5) are too fine-grained and unevenly represented for direct use in the network analyses: several categories are essentially the same biological population split by a naming artifact, and two categories are too rare and/or ambiguous to model reliably across the full age range. The following merge rules were applied:

- **Endothelial cells** $\leftarrow$ *aortic endothelial cell* + *endothelial cell of coronary artery* (same vascular endothelial lineage, split only by sampling location).
- **Myocytes** $\leftarrow$ *atrial myocyte* + *ventricular myocyte* (chamber-specific cardiomyocyte subtypes, pooled to increase per-age representation).
- **Mast cells** and **valve cells** were **excluded**: both are rare (0–32 and 0–164 cells per age point, respectively, Table 5) and, for valve cells in particular, entirely absent at four of the six age points.

All remaining ontology classes are carried forward as-is. This yields eight final cell-type categories (`final_celltype`), used throughout Sections 4–5:

#	Final category	Source cell_ontology_class label(s)
1	EndothelialCells	aortic endothelial cell; endothelial cell of coronary artery
2	Myocytes	atrial myocyte; ventricular myocyte
3	CardiacNeurons	cardiac neuron
4	EndocardialCells	endocardial cell
5	Erythrocytes	erythrocyte
6	Fibroblasts	fibroblast of cardiac tissue
7	Leukocytes	leukocyte
8	SmoothMuscleCells	smooth muscle cell

Table 7: Mapping from raw `cell_ontology_class` labels to the eight final `final_celltype` categories. Mast cells and valve cells (Table 5) are dropped.

4 Methods

4.1 Preprocessing and batch integration

Full preprocessing, QC, and exploratory results are presented in Section 3; parameters are summarized here for completeness. Cells with ≤ 200 detected genes, and genes detected in < 3 cells, were removed. Standard Seurat [4] log-normalization, scaling, and PCA were followed by Harmony batch correction across `tech` (FACS vs. droplet) [5]. Fine-grained `cell_ontology_class` labels were collapsed to eight target cell types (Section 3.7); cells without a mapped type were excluded from all four downstream analyses. All four analyses below reuse this same object, exported once (raw counts + `final_celltype/sample_id/age/sex/tech` metadata) as a single `.h5ad` file.

4.2 Gene co-expression network analysis (WGCNA/hdWGCNA)

WGCNA [1] and its single-cell extension hdWGCNA [2] were used to build a cell-type-stratified co-expression network from the pseudobulk matrix described below.

4.2.1 Pseudobulk aggregation and missing-data handling

A pseudobulk matrix (feature = `gene_celltype`, sample = mouse) was built by averaging normalized expression per mouse $\times$ cell type in linear space and re-log-transforming (avoids the bias of averaging already-logged values). Cell types missing in $> 50\%$ of mice were dropped entirely; remaining missing (mouse, cell type) entries were imputed with the same-feature mean *within age group*. Features with zero expression in every mouse, were removed.

4.2.2 Network construction

A signed, biweight-midcorrelation (`bicor`) co-expression network was built on this pseudobulk matrix with hdWGCNA’s `ConstructNetwork()` (Table 8); a signed network was chosen to keep positively and negatively co-regulated genes in separate modules. Unassigned genes (module “grey”) were excluded from every downstream step.

4.2.3 Module eigengenes, age association, and trajectory shape

Each module’s eigengene (first PC of member genes, `moduleEigengenes()`) was correlated with numeric age (Pearson, BH-FDR across modules). In parallel, each eigengene’s trajectory was fit with three nested models (null ~ 1 , linear $\sim \text{age}$, quadratic $\sim \text{age} + \text{age}^2$), compared sequentially

Table 8: Core WGCNA/hdWGCNA chosen network parameters.

<code>soft_power</code>	10
<code>corType</code> / <code>networkType</code> / <code>TOMType</code>	<code>bicor</code> / <code>signed</code> / <code>signed</code>
<code>deepSplit</code> / <code>minModuleSize</code> / <code>mergeCutHeight</code>	2 / 15 / 0.15
<code>pamStage</code> / <code>detectCutHeight</code>	TRUE / 0.995
<code>randomSeed</code>	12345

by nested F -tests; the quadratic term was kept only if it significantly improved the fit. Fitted values from the winning model were classified (15% edge tolerance on where the fitted min/max fall) into Increasing, Decreasing, U-shaped, or Inverted-U. A module was called age-associated if either the linear correlation FDR or the trajectory-model FDR was < 0.05 .

4.2.4 Hub genes and functional enrichment

Intramodular connectivity (`kME`, `signedkME`) was computed per feature; hub genes were the top-10% `kME` features within each module (a relative threshold was chosen since raw `kME` is not comparable across modules of different size/cohesion). For each module, genes were split by cell type of origin and tested for GO Biological Process, KEGG, and Reactome enrichment (`clusterProfiler` [6], `ReactomePA` [7]) against the network’s own gene universe (BH-corrected).

4.3 Cell-cell communication analysis (LIANA+)

4.3.1 Resource, method, and normalization checks

Ligand-receptor (LR) inference used LIANA+ (Python) [8] with the `mouseconsensus` resource (mouse gene symbols) and the `rank_aggregate` consensus method, which combines the magnitude/specificity scores of CellPhoneDB, Connectome, log2FC, NATMI, and SingleCellSignalR into a single

`magnitude_rank/specificity_rank`. All calls used `expr_prop = 0.1`, `use_raw = False`, and the explicit `lognorm` layer.

4.3.2 Steady-state and young-vs-old inference

LR inference was first run once on the full dataset (all ages pooled) to characterize steady-state signaling per sender–receiver cell-type pair. The identical call was then re-run separately on the youngest and the oldest age groups only, keeping all other parameters fixed.

4.3.3 Module annotation of LR pairs

Every LIANA interaction carries a `ligand_complex/receptor_complex` and a `source/target` cell type. These were joined to the WGCNA pseudobulk features, which are indexed by (gene, cell type): the ligand side on (ligand gene, sender cell type) and the receptor side on (receptor gene, receiver cell type). This directly flags which LR pairs involve a gene from a WGCNA module, and from which module, on each side of the interaction.

4.3.4 Per-mouse scoring and coverage diagnostics

For the multi-sample analysis, LIANA+’s `by_sample` function scores each (mouse, sender, receiver) combination independently, requiring a minimum number of cells (`MIN_CELLS = 10`) per (mouse, cell type) to return a score; combinations below this are left as missing. Cell coverage per mouse $\times$ cell type was tabulated and visualized beforehand to anticipate which entries would be missing.

4.3.5 Tensor construction and factorization

Per-sample LR scores (`magnitude_rank`) were assembled into a 4-dimensional tensor (mice $\times$ LR interactions $\times$ senders $\times$ receivers) with `li.multi.to_tensor_c2c()` (`how = "outer_cells"`: union of cell types across mice, intersection of LR pairs; `non_expressed_fill = 0` for LR pairs excluded by the expression-proportion filter, as opposed to genuinely missing). The fraction of remaining NaN entries in the tensor was checked (a rule of thumb of $> 40\text{--}50\%$ missing being problematic for stable rank estimation). The tensor was decomposed with TensorCell2cell [9] via `run_tensor_cell2cell_pipeline()`: non-negative CP decomposition, rank selected automatically by elbow analysis on reconstruction error (`elbow_metric = 'error'`, `smooth_elbow = FALSE`, upper bound = $\min(20, n_{\text{mice}} - 1)$), `tf_optimization = 'regular'`, `tf_init = 'random'`, `tf_svd = 'numpy_svd'`, fixed random state. The resulting factors were inspected to identify the one most correlated with the age factor.

4.3.6 Cross-validation and functional enrichment of factor loadings

The genes in the top-loading LR pairs of the age factor (top $n = 40$) were tested for hypergeometric enrichment in the age-associated WGCNA hub-gene set, against the universe of all genes appearing in any LR pair of the tensor. Separately, mouse PROGENy pathway signatures [10] (obtained in human and translated to mouse orthologs via LIANA's HCOP downloader) were regressed against the LR factor loadings with a univariate linear model (`decoupler.mt.ulm`) to assign a pathway-level interpretation to the age factor.

4.4 Differential cell-cell communication and WGCNA integration (MultiNicheNet)

4.4.1 SingleCellExperiment construction and replicate filtering

A `SingleCellExperiment` was built directly from raw counts (MultiNicheNet pseudobulks internally via `muscat/edgeR`), with `colData` explicitly aligned to `final_celltype/sample_id/age/sex/tech`. `multi_nichenet_analysis()` fits its per-cell-type pseudobulk model across *all* age levels in `colData` simultaneously; an age level backed by a single mouse cannot have its between-replicate variance estimated, which breaks DE estimation for every cell type, not only that level. Age levels with < 2 mice were therefore removed *before* any contrast was defined.

4.4.2 Contrast definition

Rather than the full set of pairwise age comparisons, a single contrast – **oldest vs. youngest** age group (the two extremes of `ordered_age_levels`) – was analyzed.

4.4.3 Prior model and DE/ligand-activity/prioritization pipeline

The mouse NicheNet v2 prior model was loaded: `lr_network`, `ligand_target_matrix`, `weighted_networks`, and `ligand_tf_matrix`. `multi_nichenet_analysis()` [12] then chains: (i) per-cell-type pseudobulk abundance/expression, (ii) per-cell-type DE via `muscat/edgeR`, (iii) NicheNet [11] ligand-activity scoring, and (iv) a multi-criteria LR prioritization table (Table 9). `sex` and `tech` were checked for confounding with `age` (cross-tabulation) and found to be almost completely confounded (several age levels represented by a single sex or a single technology); both were therefore left out as covariates/batches (`batches = NA`, `covariates = NA`) **This is a limitation of the underlying sampling design rather than a modeling choice.**

Table 9: Key `multi_nichenet_analysis()` parameters.

<code>min_cells</code>	3 (per sample $\times$ cell type)
<code>logFC_threshold</code> / <code>p_val_threshold</code>	0.50 / 0.05
<code>fraction_cutoff</code>	0.05
<code>p_val_adj</code>	FALSE (nominal p)
<code>top_n_target</code>	250

4.4.4 Diagnostics and visualization

Because `multi_nichenet_analysis()` silently drops any cell type without ≥ 2 samples per group passing `min_cells` (logged to console but not returned as a structured object), the same inclusion/exclusion table was reconstructed directly from `colData(sce)` for inspection. Cell/sample abundance balance across age groups was plotted as a QC diagnostic. For the oldest-vs-youngest contrast, a circos diagram of sender $\rightarrow$ receiver communication and an LR dotplot (combining expression, ligand activity, and logFC for the top interactions) were generated with the standard `multinichenetr` plotting functions.

4.4.5 WGCNA cross-validation

Modules significant by either module-age correlation or trajectory $FDR < 0.05$ (Section 4.2) define the “age-associated” module set. The top-250 prioritized LR pairs for the oldest-vs-youngest contrast were annotated with ligand/receptor WGCNA module membership, matched on (gene, sender cell type) for the ligand and (gene, receiver cell type) for the receptor. Overlap significance was assessed with a global hypergeometric test: of all ligand/receptor genes prioritized by MultiNicheNet that were also part of the WGCNA gene universe, how many fall in an age-associated module, versus the number expected by chance given the size of that module set and of the universe. This same overlap/hypergeometric logic was generalized to run over every contrast in `contrast_tbl` (here reducing to the single defined contrast).

4.4.6 Mechanistic follow-up: hub genes as a ligand-activity target set

Candidate (module $\times$ cell type) combinations were built by crossing module-age significance ($p < 0.05$) with which cell types actually contribute hub genes to that module, restricted to the target cell types; the top candidate (by FDR, or optionally by effect size) was selected as the focal module/receiver. That module’s hub genes, specifically from that receiver cell type, were used as the NicheNet `geneset_o1` (flagging cases with < 20 genes as statistically unstable, and skipping candidates with < 5), against a real expressed-gene background for the receiver (genes expressed in $\geq 10\%$ of its cells, not merely the WGCNA-selected HVGs, which is the statistically appropriate background for `predict_ligand_activities()`). Ligand activity was then computed two ways, as recommended by the NicheNet tutorial: **sender-focused** (separately per candidate sender cell type, restricting potential ligands to those expressed in that specific sender with a receptor expressed in the receiver) and **sender-agnostic** (any `lr_network` ligand with a receptor expressed in the receiver, regardless of source cell type). Ligands ranking highly in both are the strongest candidates; ligands unique to the sender-agnostic ranking may originate from cell types outside the modeled set and are interpreted with that caveat.

4.5 Gene regulatory network analysis (SCENIC)

Gene regulatory networks were reconstructed with the SCENIC framework [13], using its Python implementation, `pySCENIC` [14].

4.5.1 Expression matrix and TF reference

The shared raw-count export was filtered to genes expressed in $\geq 1\%$ of cells (consistent in spirit with the WGCNA HVG/expression filters) and converted to a dense cells- $\times$ -genes matrix for `arboreto`. The mouse transcription-factor list (`mm_mgi_tfs.txt`, distributed with `pySCENIC`) was intersected with this expression matrix to obtain the candidate regulator set.

4.5.2 Step 1 – GRN inference (GRNBoost2)

TF-target co-expression modules were inferred with GRNBoost2 [15] (`arboreto.algo.grnboost2`), a gradient-boosting-based regression method chosen over the original GENIE3 for its markedly lower runtime at comparable accuracy, run on a local Dask cluster (threaded scheduler). Output adjacencies (TF, target, importance) were cached to disk to allow resuming later steps without recomputation.

4.5.3 Step 2 – Regulon pruning (cisTarget)

Raw co-expression modules (`modules_from_adjacencies`) still contain many indirect/false-positive TF-target links; `cisTarget` prunes each module to targets whose regulatory regions are enriched for the TF’s binding motif, using two mouse ranking databases (500bp upstream/100bp downstream promoter-proximal, and 10kb upstream/10kb downstream distal regions; `mc_v10_clust`, RefSeq r80) together with a motif-to-TF annotation table. Motif enrichment and pruning (`prune2df + df2regulons`) produced the final regulon set, and regulon size distribution was inspected as a sanity check.

4.5.4 Step 3 – Regulon activity (AUCell) and exploration

Per-cell regulon activity was scored with AUCell (`auCell()`), the SCENIC analog of a per-cell module eigengene. Regulon activity was attached back to the annotated object, clustered (cells $\times$ regulons) to inspect global structure, and used to build a regulon-activity-based UMAP embedding (as opposed to a gene-expression-based one) – the classic SCENIC “regulon-based cell states” view. The AUCell matrix was additionally binarized per regulon (bimodal on/off calling from the AUC distribution itself, `binarize()`), analogous in spirit to WGCNA’s relative hub-gene thresholding. Regulon Specificity Score (RSS) per `final_celltype` was computed to rank, for each cell type, the regulons that best mark its identity.

4.5.5 Regulon-age correlation

A mouse-level “regulon eigengene” (mean AUCell activity per mouse, optionally computed separately per cell type) was correlated with numeric age (Pearson, BH-FDR < 0.05), exactly mirroring the WGCNA module-eigengene-age correlation (Section 4.2) so that the two outputs are directly comparable.

5 Results and discussion

5.1 Co-expression modules and age association

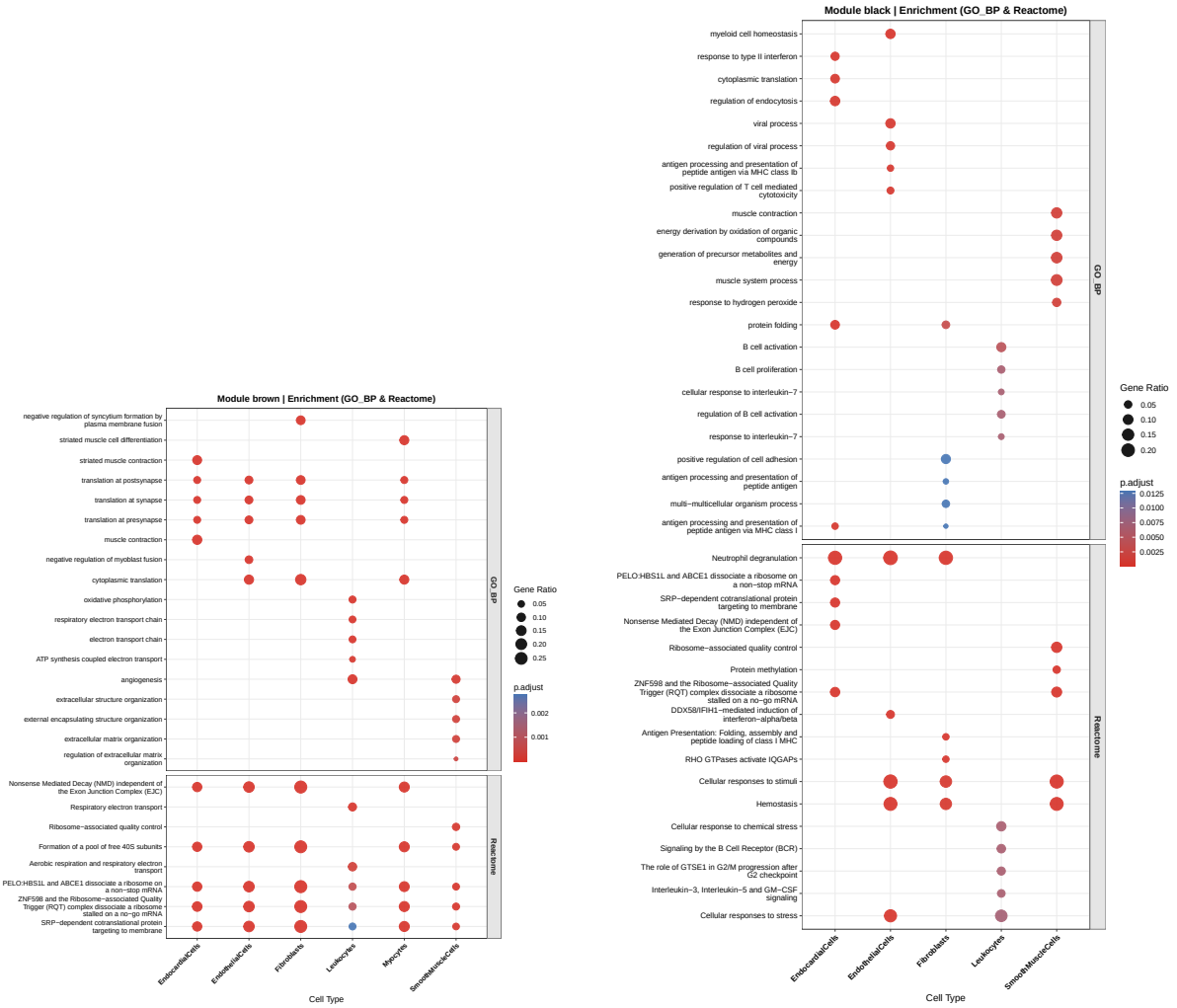
hdWGCNA identified 35 co-expression modules spanning 45,118 `gene_celltype` features (the “grey” unassigned module excluded). At nominal significance ($p < 0.05$), seven modules correlate with age (Table 10): *midnightblue* ($r = -0.54$, Leukocyte/Fibroblast/Myocyte-heavy, 1,109 features), *lightcyan* ($r = 0.55$, dominated by SmoothMuscleCells, 1,057 features), *orange* ($r = 0.50$, almost exclusively Myocytes, 586 features), *brown* ($r = 0.47$, the largest and most cell-type-balanced, 2,515 features), *tan* ($r = 0.45$, almost exclusively Myocytes, 1,482 features), *darkorange* ($r = 0.41$, SmoothMuscleCells/Myocytes-heavy, 491 features), and *yellow* ($r = -0.45$, broadly EndothelialCells/Fibroblasts, 2,488 features). None survive Benjamini–Hochberg correction across all 35 modules (minimum FDR ≈ 0.12 for midnightblue/lightcyan) – a direct consequence of the modest sample size ($n = 24$ mice) relative to the number of modules tested. Three further modules sit just outside this threshold and are discussed below for their functional signal despite not meeting it: *pink*, *greenyellow*, and *black* (all $p = 0.052$ – 0.056 , all decreasing).

Trajectory modeling adds shape information beyond the linear correlation: *orange* (Myocytes) follows a U-shaped quadratic trajectory ($p = 0.009$, net change $+0.42$), *tan* (Myocytes) an inverted-U ($p = 0.017$), *lightcyan* a further U-shaped/increasing trajectory ($p = 0.045$), while *midnightblue*, *yellow*, *brown*, and *darkorange* are each better described as monotonic (linear) trajectories.

Functional enrichment (GO BP, Reactome; per module $\times$ cell type) is markedly uneven across these seven modules. *brown* is by far the most interpretable and the least confined to a single cell type: cytoplasmic translation and ribosome/synapse-associated translation machinery are enriched across EndocardialCells, EndothelialCells, Fibroblasts, and Myocytes alike (GO BP p_{adj} down to 2.8×10^{-35}), while its Leukocyte genes instead carry an oxidative-phosphorylation/respiratory-electron-transport-chain signature and its SmoothMuscleCells genes an extracellular-matrix-organization/angiogenesis signature (Figure 10, left) – consistent with *brown* being the largest and most cell-type-balanced of the age-associated modules. *orange* (Myocytes) shows a related signal (ribosome biogenesis, protein folding, $p_{\text{adj}} < 0.04$), and *tan* shows a weaker synaptic-translation signal in its (small) Fibroblast fraction. *midnightblue* has no significant GO BP terms but two significant Reactome terms in Myocytes (post-translational protein phosphorylation; IGF transport regulation). *darkorange*’s signal is sparse and diffuse (bone growth, glomerular mesangial cell proliferation; $n = 3$ – 4 genes per term) and is not confidently interpretable given its small size. *yellow* and *lightcyan* – despite being among the age-correlated modules – returned zero significant GO BP or Reactome terms in any cell type.

Exploratory signal below the significance threshold. The three near-miss modules are functionally more informative than several modules that did clear $p < 0.05$. *black* (1,864 features, $p = 0.056$) shows the strongest and broadest enrichment of any module examined (Figure 10, right): EndocardialCells carry myeloid cell homeostasis, response to type II interferon, protein folding, and MHC-class-I antigen presentation; EndothelialCells carry viral-process response and MHC-class-Ib antigen presentation with T-cell-mediated cytotoxicity; Leukocytes carry B-cell activation/proliferation, interleukin-7 response, and B-cell-receptor signaling; and SmoothMuscleCells carry an energy-metabolism signature (oxidation of organic compounds, precursor-metabolite generation) that echoes *pink/greenyellow* below – altogether a broader innate-and-adaptive-immune, inflammaging-type signature than a single GO term can convey (all terms shown $p_{\text{adj}} < 5 \times 10^{-4}$; Neutrophil degranulation, Reactome, is shared across four of the five cell types). *pink* (1,824 features, $p = 0.053$) is enriched in Fibroblasts for

purine metabolism and aerobic respiration ($p_{\text{adj}} = 3.5 \times 10^{-4}$), and *greenyellow* (1,668 features, $p = 0.053$) shows a parallel mitochondrial signature in SmoothMuscleCells (cellular respiration, electron transport chain, $p_{\text{adj}} < 4 \times 10^{-4}$) – together suggestive of an age-related metabolic/mitochondrial shift that falls just short of the correlation threshold used here. For contrast, the single *largest* module overall, *turquoise* (8,075 features, dominated by EndothelialCells/Fibroblasts/Leukocytes), shows very strong translation-related enrichment (GO BP $p_{\text{adj}} = 4.0 \times 10^{-19}$) but essentially no age association ($p = 0.099$). A heatmap summarizing GO BP enrichment across all significant modules and cell types is shown in Figure 11.



(a) Module *brown* ($r = 0.47$, $p = 0.020$).

(b) Module *black* ($p = 0.056$, just below threshold).

Figure 10: GO BP and Reactome enrichment by cell type for the two most functionally informative modules. Point size = gene ratio, color = adjusted p -value.

Source: hdWGCNA/outputs/figures/dotplots_GO_Reactome_by_module/

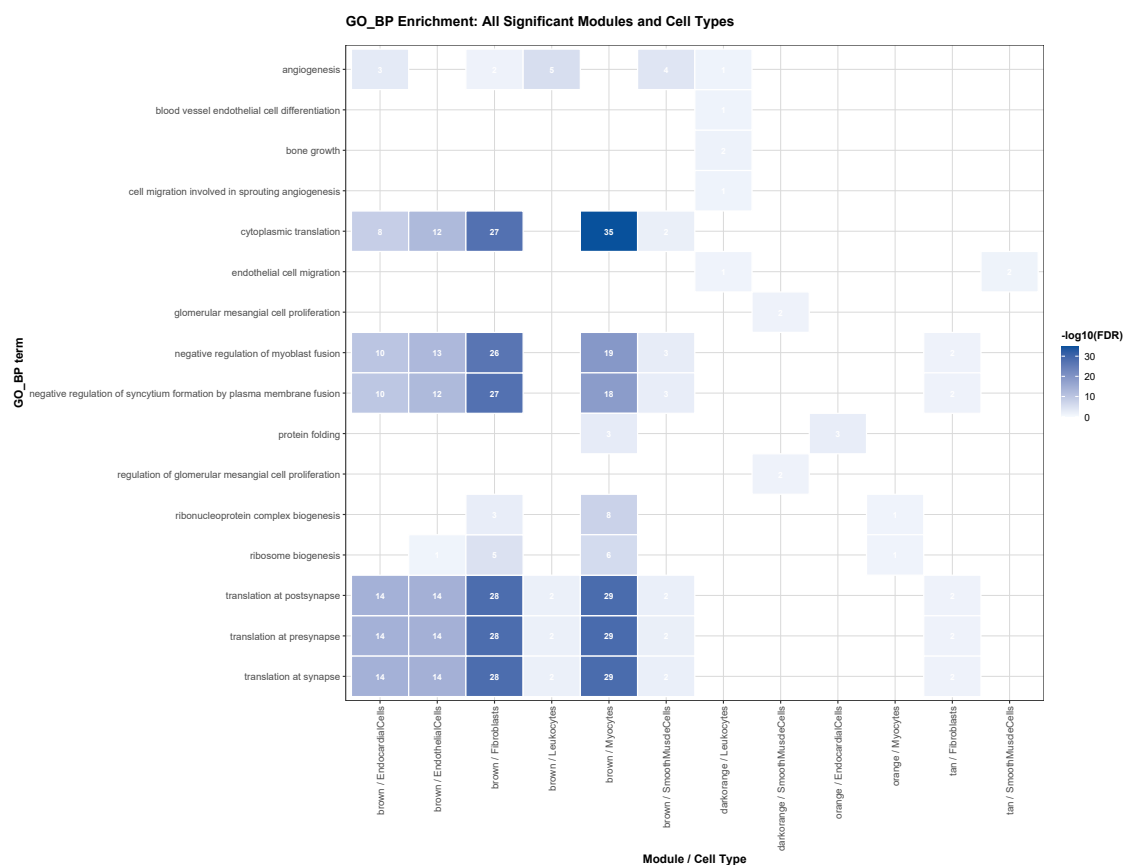


Figure 11: GO BP enrichment ($-\log_{10}\text{FDR}$, cell values = gene count) across all nominally age-associated modules and cell types.

Source: hdwgcnadata.org/outputs/figures/GO_BP_master_heatmap_all_modules.pdf

Repository Figures

Equivalent GO/Reactome dotplots and full trajectory-and-pathway figures for all 35 modules (not only the seven age-associated and three near-miss modules discussed above) are available in the repository at hdwgcnadata.org/outputs/figures/dotplots_GO_Reactome_by_module/ and hdwgcnadata.org/outputs/figures/trajectory&pathway_complete_figures_modules/, respectively (76 figures total for this layer).

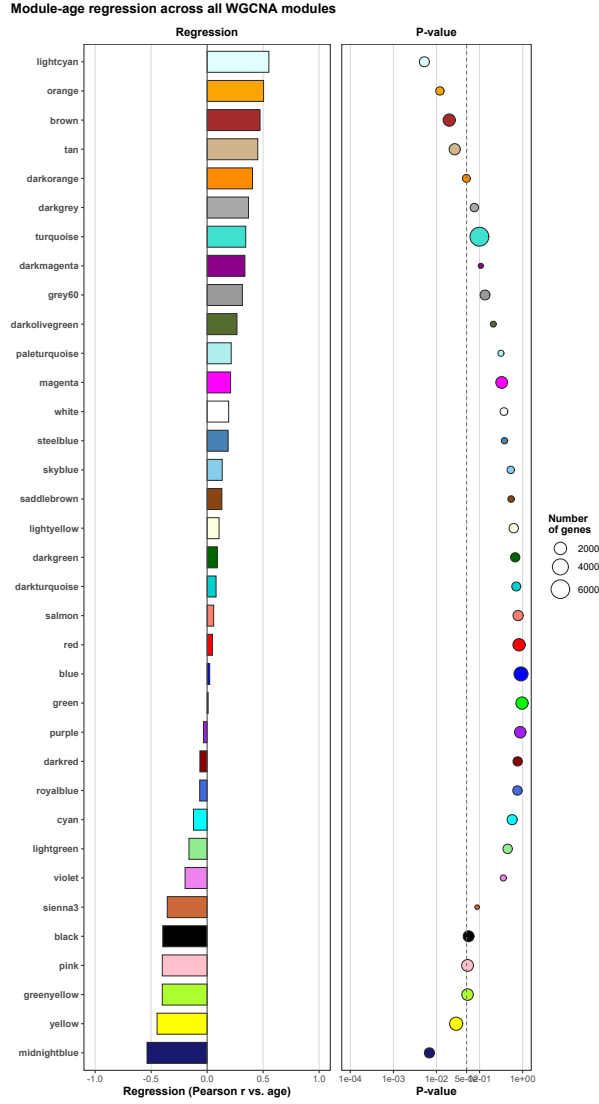


Figure 12: Module-age Pearson regression coefficient (left) and nominal p -value (right, point size = module size) for all 35 WGCNA modules, ranked by regression coefficient.

Source: hdwgcna.org/outputs/figures/figure_module_age_regression_publication.pdf

Table 10: Modules nominally associated with age ($p < 0.05$; none survive BH-FDR across all 35 modules, minimum FDR ≈ 0.12). Three further modules just below this threshold (*pink*, *greenyellow*, *black*, $p = 0.052$ – 0.056) are discussed for their functional signal.

Module	r	p	FDR	Trajectory	Size	Dominant cell type
midnightblue	−0.54	0.0069	0.12	Decreasing	1,109	Leukocytes
lightcyan	0.55	0.0052	0.12	Increasing	1,057	SmoothMuscleCells
orange	0.50	0.0120	0.14	U-shaped	586	Myocytes
brown	0.47	0.0198	0.17	Increasing	2,515	Myocytes/mixed
tan	0.45	0.0265	0.17	Inv. U	1,482	Myocytes
darkorange	0.41	0.0495	0.19	Increasing	491	SmoothMuscleCells/Myocytes
yellow	−0.45	0.0286	0.17	Decreasing	2,488	EndothelialCells

5.2 Cell-cell communication (LIANA+)

Filtering LIANA+’s consensus ligand-receptor output to interactions where at least one side maps onto the WGCNA gene universe leaves 4,331 LR pairs; of these, 396 have both ligand and receptor sitting in an age-associated WGCNA module. *SmoothMuscleCells*, *Myocytes*, and *Fibroblasts* are the most frequent senders (1,136 / 789 / 738 interactions), while *SmoothMuscleCells*, *Myocytes*, and *EndocardialCells* are the most frequent receivers. The single highest-confidence steady-state interaction overall is *Nppa* → *Npr3* (Myocytes → EndocardialCells), with both ligand and receptor in the *brown* module.

Multi-sample tensor decomposition (rank 8, elbow-selected) covered 21 of the 24 mice (three excluded for insufficient per-mouse cell-type coverage at MIN_CELLS = 10). Contrary to its designation as “the age factor,” Factor 1’s per-mouse loading correlates only weakly with age in this subset ($r = -0.07$, $n = 21$); of the eight factors, Factor 2 shows the strongest (still non-significant) age correlation ($r = -0.35$), followed by Factor 4 ($r = 0.27$). Factor 1’s sender loadings are broadly distributed across all six cell types (0.30–0.47), but its receiver loading is almost entirely concentrated on Leukocytes (0.9999 vs. ≤ 0.002 for every other cell type) – i.e., regardless of its age correlation, Factor 1 isolates a coherent *leukocyte-directed* signaling axis. Consistent with this, its top-loading LR pairs are dominated by innate-immune/complement signaling: *Rps19* → *C5ar1*, *Gnai2* → *Ccr5*, *Apoe* → *Trem2*, *Cfh* → *Itgam*, and *Tnfsf12* → *Cd163*; only 2 of the top 13 pairs hit a WGCNA hub gene (*Ptprc*, *Cfh*). PROGENy pathway scoring of Factor 1’s loadings returns a single nominally significant pathway, p53 ($p = 0.038$); by contrast, Factor 4 – the second-most age-correlated factor – shows several nominally significant pathways (EGFR, Estrogen, MAPK, NF κ B, PI3K, TNF α ; all $p < 0.05$).

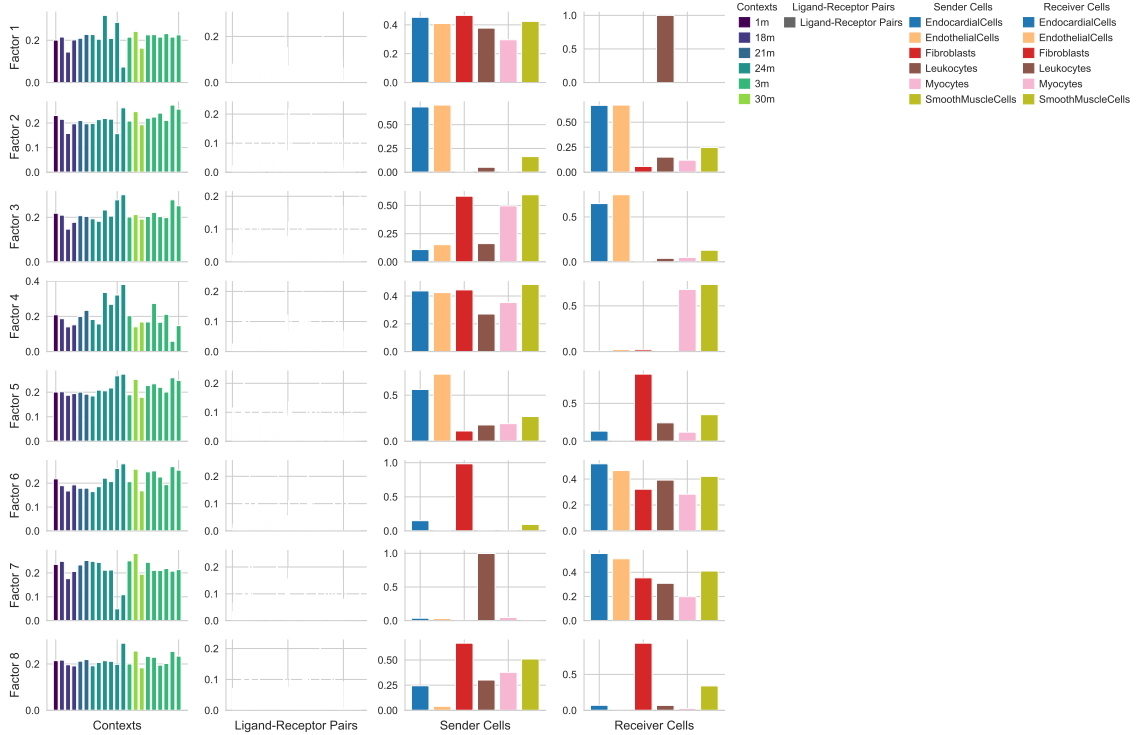


Figure 13: Tensor-cell2cell decomposition, all 8 factors: per-mouse loading (Contexts, colored by age), LR-pair loadings, and sender/receiver cell-type loadings. Factor 1’s receiver loading is visibly concentrated almost entirely on Leukocytes (bottom-right panel, row 1).

Source: [LIANA+&tensor2cell/output/plots/01_tensor_factors_plot.pdf](#)

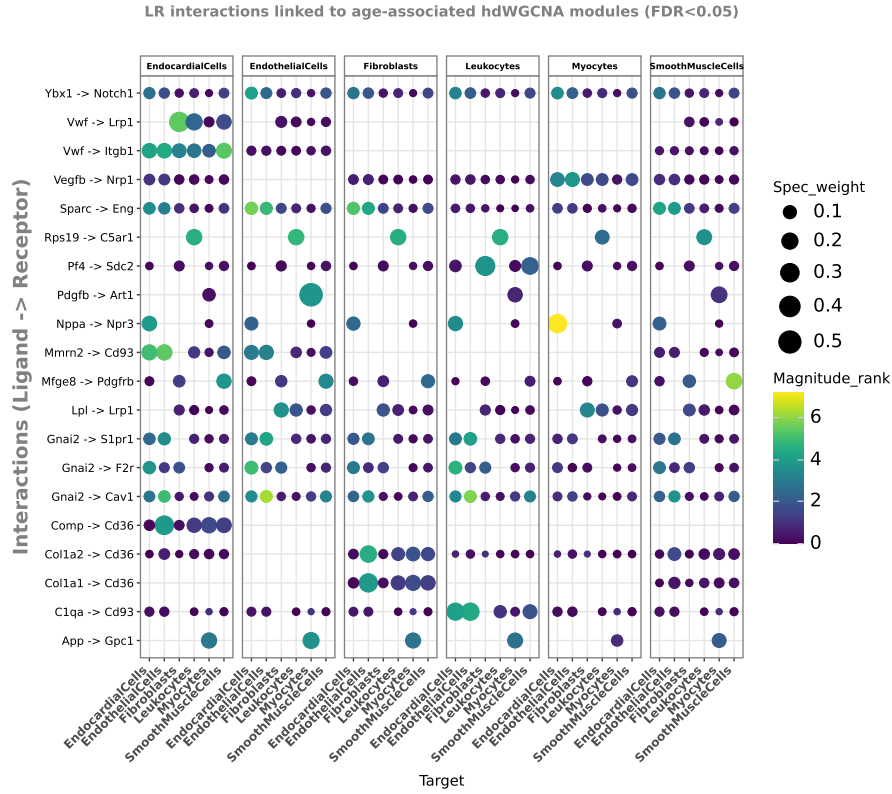


Figure 14: LR interactions with both ligand and receptor mapping onto an age-associated WGCNA module (FDR < 0.05), faceted by sender (top) and receiver (x-axis). Point size = specificity weight, color = magnitude rank.

Source: [LIANA+&tensor2cell/output/plots/liana_dotplot_hdwgcn.pdf](#)

Repository Figures

Further LIANA+ figures – per-cell-type sender dotplots, per-module hub-gene kME plots, young-vs-old heatmaps/scatterplots, and PROGENy bar/heatmaps for all 8 factors (29 further figures) – are available in the repository at [LIANA+&tensor2cell/output/plots/](#) and [LIANA+&tensor2cell/output/GSEA/](#).

5.3 Differential communication and cross-validation (MultiNicheNet)

The oldest-vs-youngest contrast (30m vs. 3m) was evaluated across 8,514 sender–receiver–ligand–receptor combinations; the top 250 prioritized interactions (95 unique ligands, 179 unique LR pairs) were carried forward for cross-validation. A striking pattern in the golden table: all 250 top-prioritized interactions favor the 30-month (old) group – none of the top interactions were prioritized as higher in young mice – consistent with a net *gain* of intercellular signaling with age in this dataset rather than a balanced mix of gains and losses. The signature is dominated by the TGF β ligand family (*Tgfb1*, *Tgfb2*, *Tgfb3*) signaling from *Leukocytes* and *Fibroblasts* to receptors on *Fibroblasts*, *EndothelialCells*, and *Leukocytes* (*Tgfb1/2/3*, *Acvr1*, *Eng*, *Sdc2*, *Pdgfra*), alongside complement (*Cfh*–*Cfb*), collagen/integrin (*Postn*–*Itgav*/*Itgb5*), and BMP (*Bmp2/4*–*Acvr1*/*Bmpr1a*) signaling.

Cross-validated against the WGCNA age-associated module set: 89 of the 250 top interactions (35.6%) have their ligand and/or receptor in an age-associated module, and 11 (4.4%) have both sides in a module – a global hypergeometric enrichment of $p = 1.6 \times 10^{-4}$ (Figure 16), i.e. this overlap is far larger than expected by chance. It is overwhelmingly attributable to the *brown* module (66/73 ligand hits, 25/27 receptor hits) – the same module that is independently the strongest, largest age-*increasing* co-expression module (Section 5.1) and the module underlying the top LIANA+ steady-state interaction (Section 5.2). This three-way convergence on *brown* is the strongest cross-validated signal in the current analysis (Section 5.5).

For the mechanistic ligand-activity follow-up (hub genes of an age-associated module, per cell type, as the NicheNet target set), the top candidate upstream ligands were modest in strength (AUPR 0.002–0.05, i.e., close to what would be expected by chance for a random gene set of this size): *lightcyan*’s SmoothMuscleCells hub genes point to **Thy1** (AUPR 0.0021); *midnightblue*’s Leukocyte hub genes point to **Pf4** (AUPR 0.0054) and its Myocyte hub genes to **Ahsg** (AUPR 0.044, the strongest of the candidates examined); *lightcyan*’s EndocardialCells hub genes point to **Apoe** (AUPR 0.046). These should be treated as weak, exploratory candidates rather than confident mechanistic hypotheses given the low AUPR values.

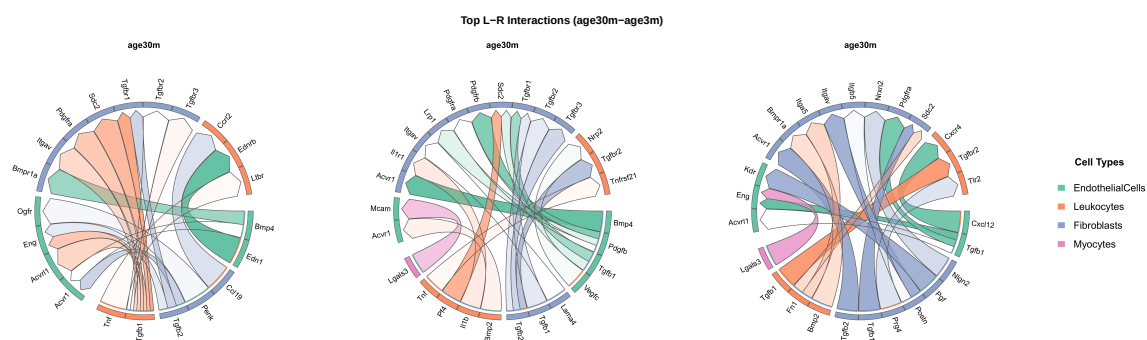


Figure 15: Top-50 prioritized sender→receiver ligand–receptor interactions for the oldest-vs-youngest (age30m–age3m) contrast, split into three panels for readability. TGF β /BMP/complement family ligands (*Tgfb1/2/3*, *Bmp4*, *Pdgfb*) and their receptors (*Tgfr1/2*, *Bmpr1a*, *Acr1*, *Itgav*, *Eng*) recur across all three panels.

Source: MultiNicheNet/output/plots/plot_circos_3panels_top50.pdf

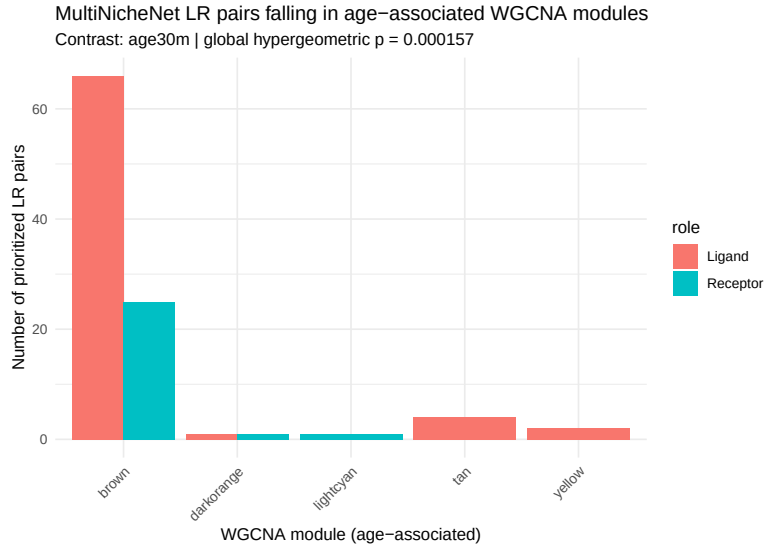


Figure 16: Number of MultiNicheNet-prioritized ligands/receptors falling in each age-associated WGCNA module (global hypergeometric $p = 1.6 \times 10^{-4}$).

Source: [MultiNicheNet/output/plots/plot_LR_wgcna_module_overlap.pdf](#)

Repository Figures

Further MultiNicheNet figures – per-(module, cell type) ligand-activity, ligand-receptor, and ligand-target heatmaps (54 figures), per-(module, cell type) signaling-path diagrams (21 figures), QC/abundance diagnostics, and the full ultra-wide LR dotplot (3 parts) – are available in the repository at [MultiNicheNet/output/plots/](#)

5.4 Gene regulatory networks (SCENIC)

As described in Section 4.5, only Step 1 (GRNBoost2 TF-target inference) completed; its output (adjacencies.zip, raw TF-target importance scores) is available in the repository, but no regulons, AUCell activity matrix, RSS table, or regulon-age correlations were produced.

5.5 Cross-method integration summary

Across the three completed analytical layers, the *brown* module is the strongest convergent signal: it is (i) the largest, most cell-type-balanced age-increasing WGCNA module (nominal $p = 0.020$, largely driven by translation/ribosome-biogenesis biology in Myocytes), (ii) the module underlying the single top-ranked LIANA+ steady-state interaction (Nppa–Npr3, Myocytes→EndocardialCells), and (iii) by far the dominant module among MultiNicheNet’s top-250 age-prioritized interactions (66 ligand hits, 25 receptor hits, overwhelmingly TGF β /BMP/complement signaling from Leukocytes and Fibroblasts). Together, WGCNA and MultiNicheNet independently converge on a TGF β -complement-collagen fibro-inflammatory axis, sourced predominantly from Leukocytes and Fibroblasts and directed broadly across cell types, with *all* top differential interactions gaining strength with age.

6 Limitations

Statistical Power and Multiple-Testing Constraints in WGCNA

No co-expression module derived from the WGCNA framework survived multiple-testing correction. Although seven candidate age-associated modules (Section 5.1) demonstrated nominal significance at $p < 0.05$, their false discovery rates remained above standard thresholds ($\text{FDR} \geq 0.12$). Furthermore, three additional modules (*pink*, *greenyellow*, and *black*) exhibited marginal trends just outside this threshold ($0.052 \leq p \leq 0.056$) despite harboring strong biological signals. With a sample size of $n = 24$ mice evaluated across $m = 35$ identified modules, the study is statistically underpowered.

Tensor Factor Selection Criteria and Sample Coverage

Factor selection within the LIANA+ tensor decomposition framework was conducted positionally rather than being dynamically data-driven. Factor 1 was designated *a priori* as the primary “age factor”; however, its correlation with age ($r = -0.07$, across $n = 21$ mice with sufficient coverage) proved substantially weaker than that of Factor 2 ($r = -0.35$) or Factor 4 ($r = 0.27$). Furthermore, only 21 of the 24 mice met the cell-count threshold ($\text{MIN_CELLS} = 10$) necessary for inclusion in the tensor analysis, reducing sample coverage.

Confounding Variables and Temporal Resolution in MultiNicheNet

The experimental sampling design exhibits unavoidable confounding between age and both sex (**sex**) and technical batch effects (**tech**). Consequently, the MultiNicheNet differential expression framework cannot unequivocally disentangle age-driven cell communication changes from background technical or sex-specific variations in the extreme contrast between oldest and youngest cohorts. Additionally, because MultiNicheNet analyzes a single binary contrast (oldest vs. youngest), it fails to capture non-linear or intermediate temporal dynamics (which are only partially captured by the continuous trajectory models in WGCNA and LIANA+ tensor factors).

7 Software and Reproducibility

The computational framework integrates R (version ≥ 4.2) for co-expression networks and differential communication with Python (version ≥ 3.9) for cell-cell communication tensor decomposition and single-cell gene regulatory network inference (Table 11). To prevent environment instability, execution was partitioned into isolated environments; specifically, LIANA+/Tensor-cell2cell and pySCENIC were maintained in separate Python stacks due to strict version dependencies (notably involving `ctxcore`, `dask`, and `distributed`).

Full execution of the workflow requires external prior knowledge resources, including the NicheNet v2 mouse ligand-receptor and ligand-target models, mouse transcription factor annotations (`mm_mgi_tfs.txt`), and the mouse *cisTarget* ranking databases (`mm10 500bp/10kbp` feather files with motif annotations). To ensure computational reproducibility, random seeds were fixed across stochastic operations, including WGCNA network construction, UMAP layout projections, and tensor factorization runs. All analytical layers consume a single, unified scRNA-seq reference object sharing identical cell-type annotations (`final_celltype`) and animal metadata (`sample_id`), permitting direct cross-validation of targets across methods without gene or cell-type re-mapping.

Random seeds were fixed where supported (WGCNA network construction, UMAP layouts, tensor factorization).

Table 11: Core software tools utilized by analysis layer.

Analysis Layer	Software / Packages	Language
Preprocessing & Integration	Seurat, harmony	R
Co-expression Networks	WGCNA, hdWGCNA	R
Cell-Cell Communication	LIANA+, cell12cell / Tensor-cell12cell, decoupler	Python
Differential Communication	multinichenetr, nichenetr	R
Gene Regulatory Networks	pySCENIC, arboreto (GRNBoost2), AUCell	Python / R
Functional Enrichment	clusterProfiler, ReactomePA, org.Mm.eg.db	R

8 Data and code availability

Raw single-cell RNA-seq data (FACS and droplet, mouse heart) are publicly available from the *Tabula Muris Senis* consortium [3]. The NicheNet/MultiNicheNet mouse prior models (ligand-receptor network, ligand-target matrix, weighted networks, ligand-TF matrix) and the SCENIC mouse cisTarget ranking databases and motif annotation table are publicly available from their respective repositories (Zenodo and <https://resources.aertslab.org/>, respectively).

All analysis code, together with the intermediate and final output tables/figures generated for this report, is organized in four top-level folders, one per analysis layer (hdWGCNA/, LIANA+&tensor2cell/, MultiNicheNet/, SCENIC/), and is available at:

<https://github.com/Olaia1/aging-heart-multiomic-scrna>

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